



THE UNIVERSITY OF
MELBOURNE

Marketing Research

Week 8





Housekeeping

- Distribute the survey – reach out if you have any concerns about the comments from instructors
- Quiz #4 will be available this Thursday
- Discussion boards on Canvas
 - One for general questions
 - One for the group project
- Deliverables for group project
 - Write-up
 - Slide deck



Group Project: Write-up (≤ 6000 words)

TITLE PAGE (NOT INCLUDED IN WORD COUNT)

TABLE OF CONTENTS (NOT INCLUDED IN WORDCOUNT)

EXECUTIVE SUMMARY (~200 words)

- Research objectives
- Main findings/results
- Main conclusions and recommendations

INTRODUCTION

- Motivate the research project and outline your main research questions.

LITERATURE REVIEW & SECONDARY DATA (~1000 words)

- literature research and secondary data related to your research questions



Group Project: Write-up (cont)

QUALITATIVE RESEARCH (~2000 words)

- Follow the qualitative data analysis process and generate key themes of finding.
- Summarise key insights from the qualitative data analysis.
- Insert KEY graphs, tables, figures, or pictures, if there are any, in the main text (not part of the word count)
- Appendix for
 - Other graphs, tables, pictures, etc
 - Interview guide/focus group guide/ethnography script used for the qualitative research in the appendix. DO NOT attach interview transcripts.
 - If you decide to use other qualitative research methods other than the three listed above, please also include a description of the procedure in the appendix.
- Keep evidence of your data collection, e.g., interview recordings, and I may request such evidence in the process of assessing your final report.



Group Project: Write-up (cont)

QUESTIONNAIRE DESIGN

- Explain the design of the questionnaire (e.g., how the questionnaire is structured, the purpose of each section). Label the purpose of questions in the questionnaire (e.g., Q1 – Research question 1; Q25 – Classification/Marketing Segmentation).
- Include the (Qualtrics) survey link.
- Include the Word document of the (Qualtrics) questionnaire in the Appendix (Questionnaire is not part of the word count).



Group Project: Write-up (cont)

ANALYSIS & RESULTS (~2000 words)

- Provide a demographic profile of your sample using the descriptive statistics and visual presentations when possible.
- Provide a summary table listing at least six hypotheses (see below), corresponding research questions, and corresponding measurement scales, and briefly justify the hypotheses by linking to findings from the exploratory research.
 - H1: to be tested with Chi-square
 - H2: to be tested with t-test
 - H3: to be tested with t-test (different type of t-test from H2)
 - H4: to be tested with one-way ANOVA
 - H5: to be tested with multiple regression
 - H6: to be tested with any analysis technique
- Describe the results for each test. Include output tables in the main text and highlight the important p-values. These materials are not part of the word count. Other graphs, tables, pictures, etc, can be placed in the appendices.



Group Project: Write-up (cont)

CONCLUSIONS AND RECOMMENDATIONS

- Integrate literature review, secondary data, and qualitative and quantitative findings for each research question
- Provide recommendations based on research findings.

APPENDICES (NOT INCLUDED IN WORD COUNT)

LIST OF REFERENCES (NOT INCLUDED IN WORD COUNT)



Group Project: Slide Deck

PAGE 1

- Project title
- One sentence highlight
- Team number/names
- Who contributes to what? (e.g., the whole team splits every step of work evenly; Student A conducts statistical analyses; Student B works on analysing qualitative data)

PAGE 2

- Detailed research questions
- Key secondary data research findings

PAGE 3

- Key qualitative research findings

PAGE 4-5

- Key quantitative research findings

PAGES 6

- Key recommendations (based on data)



Recap: Testing for Differences

Type of Tests	Purpose of Test
One Sample T-Test	Compare the mean of a question answered by one group of subjects against a constant on metric data
Paired Samples T-Test	Compare two means of two different questions answered by the same group of subjects on metric data
Independent Samples T-Test	Compare the mean of a question answered by two different groups of subjects on metric data
ANOVA	Compare the means of a question answered by more than two different groups on metric data



Topics for today

1. Recap testing for Difference: t-tests
2. Testing for Difference: (one-way) ANOVA
3. Activities on SPSS



Which type of t-test is it?

Scenario 1: Blackmores compares first-time buyers vs. repeat buyers in terms of trust scores (1–10 scale). Each group has 150 respondents. The brand hypothesizes that repeat buyers report significantly higher trust than first-time buyers.

Scenario 2: Swisses wants to test if their new turmeric supplement meets the industry benchmark satisfaction score of 6.5 (on a 10-point scale). They surveyed 120 recent users. The hypothesis is that the average user satisfaction is significantly higher than 6.5.

Scenario 3: Nature's Way hypothesizes that exposure to an influencer campaign significantly shifts the intention to buy. The same respondents rate purchase intent before and after viewing the campaign (1-7 scale).

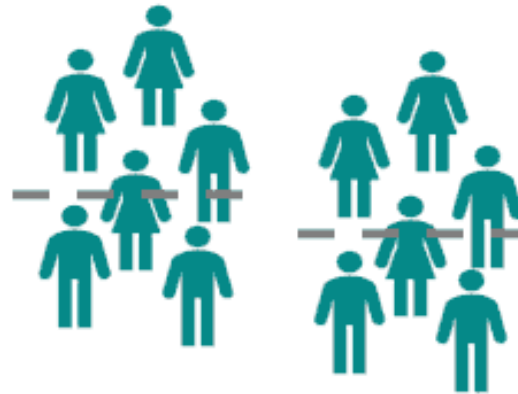
Testing for Differences (t-tests)

1.



One sample
t-test

2.



Independent
samples t-test

3.



Paired
samples t-test



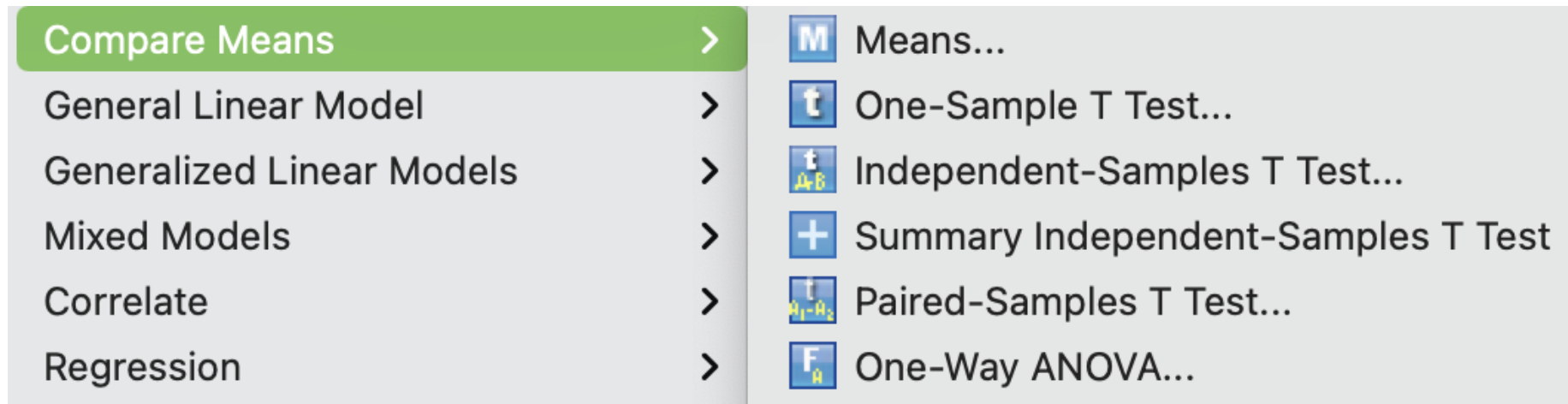
Test for Differences in SPSS

All the t-tests for population means:

- Analyze → Compare Means →
 - One Sample T Test
 - Independent-Samples T Test
 - Paired-Samples T Test
 - One-Way ANOVA

The two-way ANOVA

Analyze → General Linear Model → Univariate





Independent Sample t-Test: SPSS Output

	Levene's Test for Equality of Variances		t-test for Equality of Means			
	F	Sig.	t	df	Two-sided p	Mean Difference
Equal variances assumed		p-value 1	If p-value 1 ≥ 0.05			
Equal variances not assumed			If p-value 1 < 0.05			

There are 3 p-values in the table; one for Levene's test, and the other two for t-test

Why two p-values for the t-test?

- The t-statistics depend on whether the equal-variances assumption holds or not.



Independent Samples t-Test Cheat Sheet (Levene's Test → Independent Samples t-Test)

1. Step 1. Levene's Test: Test for equality of variances

- a) This is to test the “equal variances” assumption. The p-value in the Independent Samples t-Test depends on whether this assumption is satisfied.
- b) Null Hypothesis: $H_0: \sigma_1^2 = \sigma_2^2$;
- c) Alternative Hypothesis: $H_0: \sigma_1^2 \neq \sigma_2^2$

2. Step 2. Main test: Independent Samples t-Test

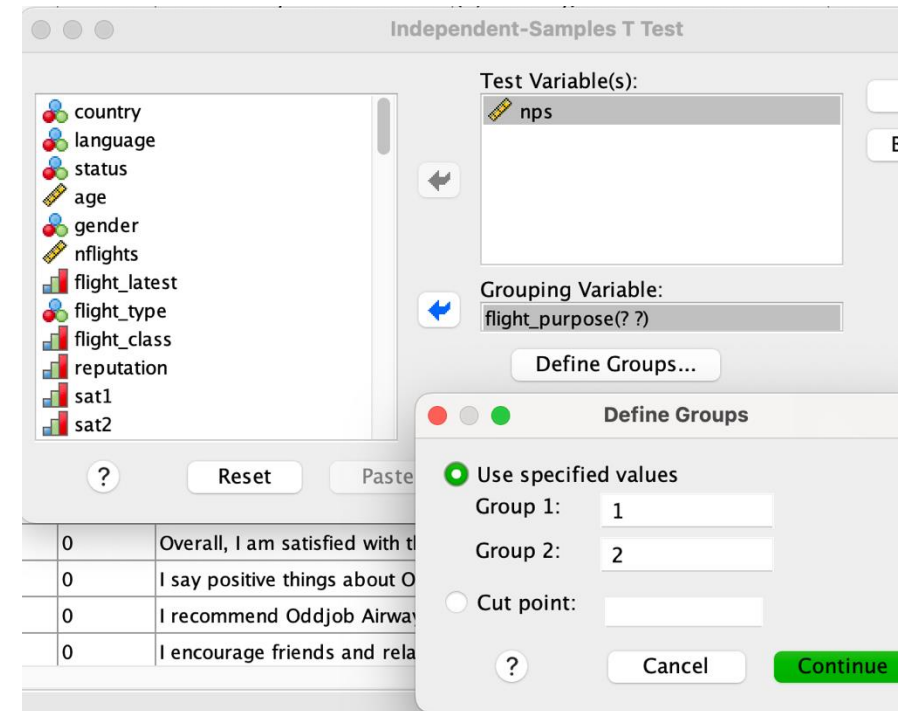
- a) Null Hypothesis: $H_0: \mu_1 = \mu_2$; Alternative Hypothesis: $H_1: \mu_1 \neq \mu_2$
- b) Use the p-value based on whether equal variances assumed or not
- c) Statistical decision using p-value
- d) Interpretation (e.g., is one average higher than the other one?)



Example

H1: There is a difference in the likelihood to recommend the Oddjob Airways (NPS) between different flight purpose (business vs. private). We hypothesize that the those who fly for business are more likely to recommend Oddjob Airways to a friend or colleague.

- Statistical test: independent samples t-test
- Null hypothesis: $\mu_{business} = \mu_{private}$, where μ = population mean of NPS.
- Alternative hypothesis: $\mu_{business} \neq \mu_{private}$



SPSS output for the independent samples t - test

Group Statistics					
	What was the purpose of your latest trip?	N	Mean	Std. Deviation	Std. Error Mean
How likely is it that you would recommend our company to a friend or colleague?	Business	525	8.01	2.432	.106
	Private	540	8.55	2.568	.111

Independent Samples Test							
Levene's Test for Equality of Variances				t-test for Equality of Means			
	F	Sig.	t	df	One-Sided p	Two-Sided p	Mean Difference
Equal variances assumed	1.560	.212	-3.537	1063	<.001	<.001	-.542
Equal variances not assumed			-3.539	1062.268	<.001	<.001	-.542

- Sample means: $\bar{x}_{business} = 8.01$; $\bar{x}_{private} = 8.55$
- P-value for Levene's Test for Equality of Variances = 0.212 > 0.05
- P-value for the t-test < 0.001 < 0.05 → Reject the null hypothesis of the t-test → $\mu_{business} \neq \mu_{private}$.
- Based on the sample means, data support $NPS_{business} < NPS_{private}$.



ANOVA

❑ Compare the means of *three or more groups*

Assumption

- The dependent variable is measured on an interval or ratio scale.
- The observations are independent.

Examples

- Do light, medium, and heavy users differ in their monthly disposable income?
- Do customers across four segments differ in their attitude towards a certain brand?
- Is there a significant difference in hours spent on the internet daily among the users of TikTok, Instagram, and Rednote?



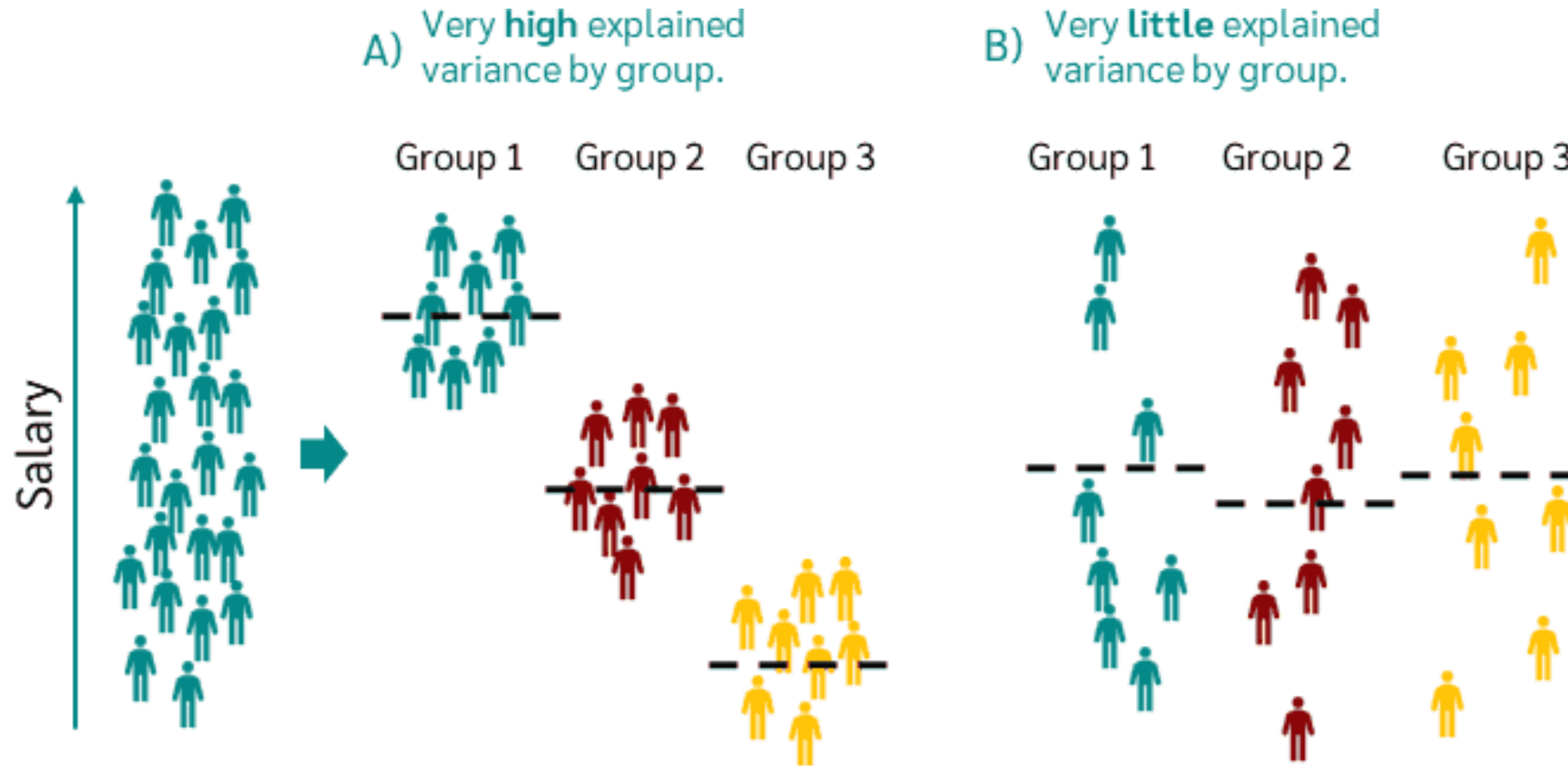
Why ANOVA?

- When comparing 3+ groups, you might think to do several t-tests between pairs.
- But this inflates the Type I error rate (false positives).

Example:

- Suppose you're comparing 4 customer segments (e.g., Gen Z, Millennials, Gen X, Boomers) on supplement satisfaction.
- The number of comparisons grows rapidly. There are 6 pairwise comparisons ($4C2 = 6$).
- Each t-test at $\alpha = 0.05$ has a 5% chance of error. The overall error rate increases with each test:
$$\text{familywise Error Rate} = 1 - (1 - \alpha)^k = 1 - (0.95)^6 = 0.265$$
- So, 26.5% chance of making at least one false discovery!

One-way ANOVA: Illustration





Types of ANOVA

- **One-way ANOVA:** Compare means across 3 or more groups based on one independent variable (factor).
 - Tests if group means differ significantly.
 - Uses one categorical IV. For instance, age group, region, or user type.
 - Example in DS Research: Compare satisfaction with a new supplement across 3 age groups: Young adults, middle-aged, and seniors.
 - Assumptions:
 1. Normal distribution within groups;
 2. Homogeneity of variances;
 3. Independence of observations
- **Two-way ANOVA:** Examine the effect of two independent variables (factors) on one outcome, and whether they interact. The assumptions are the same as in One-Way ANOVA
 - Example: Factor 1 = age group; Factor 2 = gender; **Interaction effect** = does **age** influence satisfaction **differently for men and women?**

One-way ANOVA

$$H_0: \mu_1 = \mu_2 = \mu_3$$

H_1 : At least two of μ_1 , μ_2 , and μ_3 are different

The total variation in the data (indicated by SS_T) is separated into two parts:

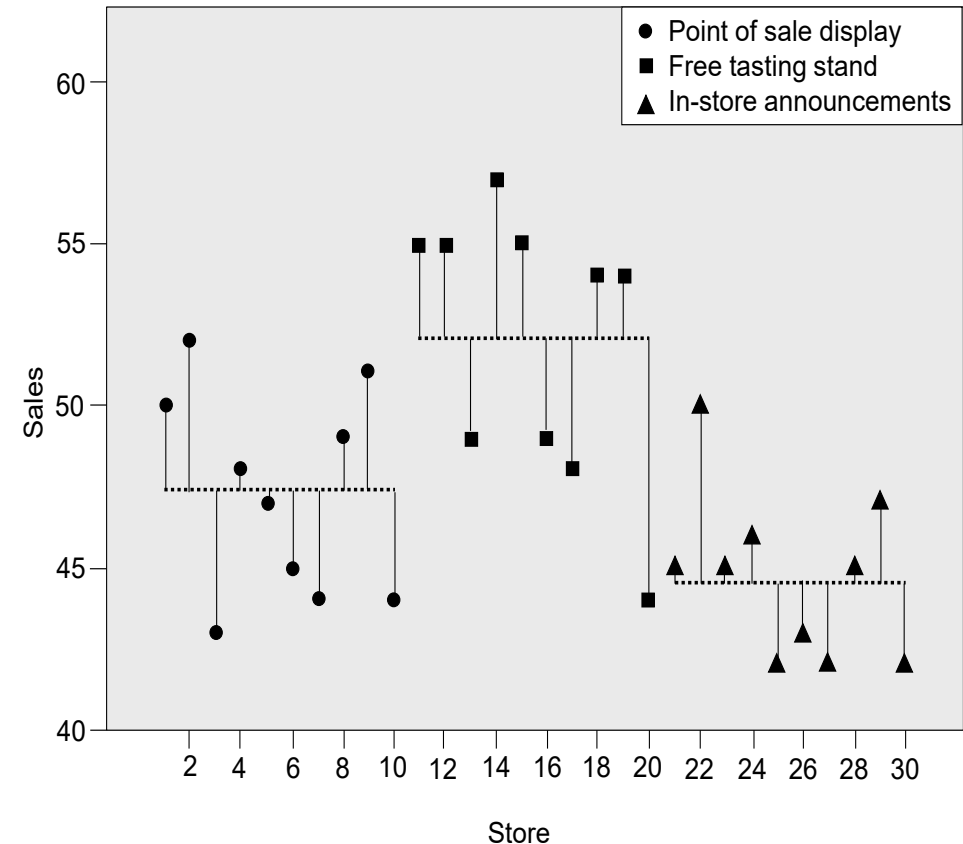
(1) the between-group variation (SS_B) (also “explained variation”), and

(2) the within-group variation (SS_W) (also called “unexplained variation”/ “(random) noise”).

$$SS_T = \sum_{i=1}^N (X_i - \bar{X})^2$$

$$SS_B = \sum_{j=1}^K n_j (\bar{X}_j - \bar{X})^2 \quad SS_W = \sum_{j=1}^K \sum_{i=1}^{n_j} (X_{ij} - \bar{X}_j)^2$$

A high proportion of *between-group variation* indicates a systematic influence of the group variable (type of promotion; independent variable) on the dependent variable (sales)



One-way ANOVA

Example

status	Numeric	8	0	Traveler Status	{1, Blue}...
overall_sat	Numeric	8	0	Overall, I am satisfied with the price performance ratio of Oddjob Airways.	{1, Fully dis...}

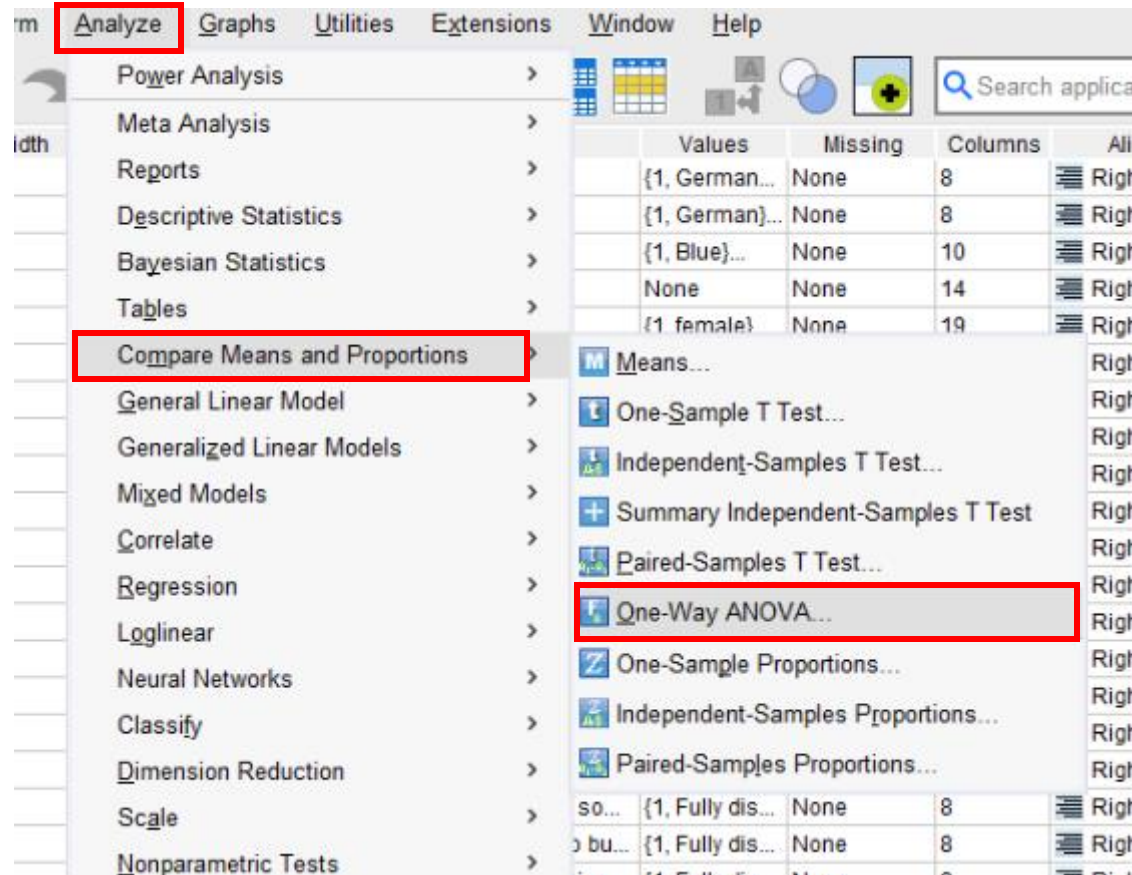
Value ▾	Label
1	Blue
2	Silver
3	Gold

Value ▾	Label
1	Fully disagree
7	Fully agree

- Three groups; one interval scale outcome variable -> One-way ANOVA
- Significance level: 5%
- Factor (grouping variable): Traveler Status (Status)
- Testing variable: Overall satisfaction (overall_sat)
- Hypothesis: At least one traveler status group (Blue, Silver, or Gold) significantly differs in their overall satisfaction with the price-performance ratio of Oddjob Airways.

One-way ANOVA

Analyze → Compare Means and Proportions One-way ANOVA → Dependent List = DV → Factor = IV → [Options] → Descriptive and Homogeneity of variance test → continue





One-way ANOVA

Analyze → Compare Means and Proportions One-way ANOVA → Dependent List = DV → Factor = IV → [Options] → Descriptive and Homogeneity of variance test → continue

The screenshot shows the SPSS One-Way ANOVA dialog box and its Options sub-dialog. In the main dialog, 'overall_sat' is selected in the Dependent List and 'status' is selected in the Factor list. The 'Options...' button is highlighted. The Options sub-dialog is open, showing the 'Statistics' section with 'Descriptive' and 'Homogeneity of variance test' checked. The 'Confidence Intervals' section shows a level of 0.95. The 'Continue' button is highlighted.

Country	Language	Age	Gender	nflights	flight_latest	flight_type	flight_purpose	flight_class	nps	reputation	sat1	sat2
0	Oddjob Airways is a reputable airline.	{1, Fully dis...	None	8	Right	Or						
0	In general, I am satisfied with Oddjob Airw...	{1, Fully dis...	None	8	Right	Or						
0	Oddjob Airways is very close to my ideal...	{1, Fully dis...	None	8	Right	Or						



One-way ANOVA

Descriptives

Overall, I am satisfied with the price performance ratio of Oddjob Airways.

	N	Mean	Std. Deviation	Std. Error	95% Confidence Interval for Mean		Minimum	Maximum
					Lower Bound	Upper Bound		
Blue	677	4.47	1.641	.063	4.35	4.60	1	7
Silver	245	4.03	1.560	.100	3.84	4.23	1	7
Gold	143	3.99	1.556	.130	3.73	4.24	1	7
Total	1065	4.31	1.625	.050	4.21	4.40	1	7

Tests of Homogeneity of Variances

		Levene Statistic	df1	df2	Sig.
Overall, I am satisfied with the price performance ratio of Oddjob Airways.	Based on Mean	.907	2	1062	.404
	Based on Median	.068	2	1062	.934
	Based on Median and with adjusted df	.068	2	1017.925	.934
	Based on trimmed mean	.771	2	1062	.463

p-value is above 0.05
→ equal variances assumed

ANOVA

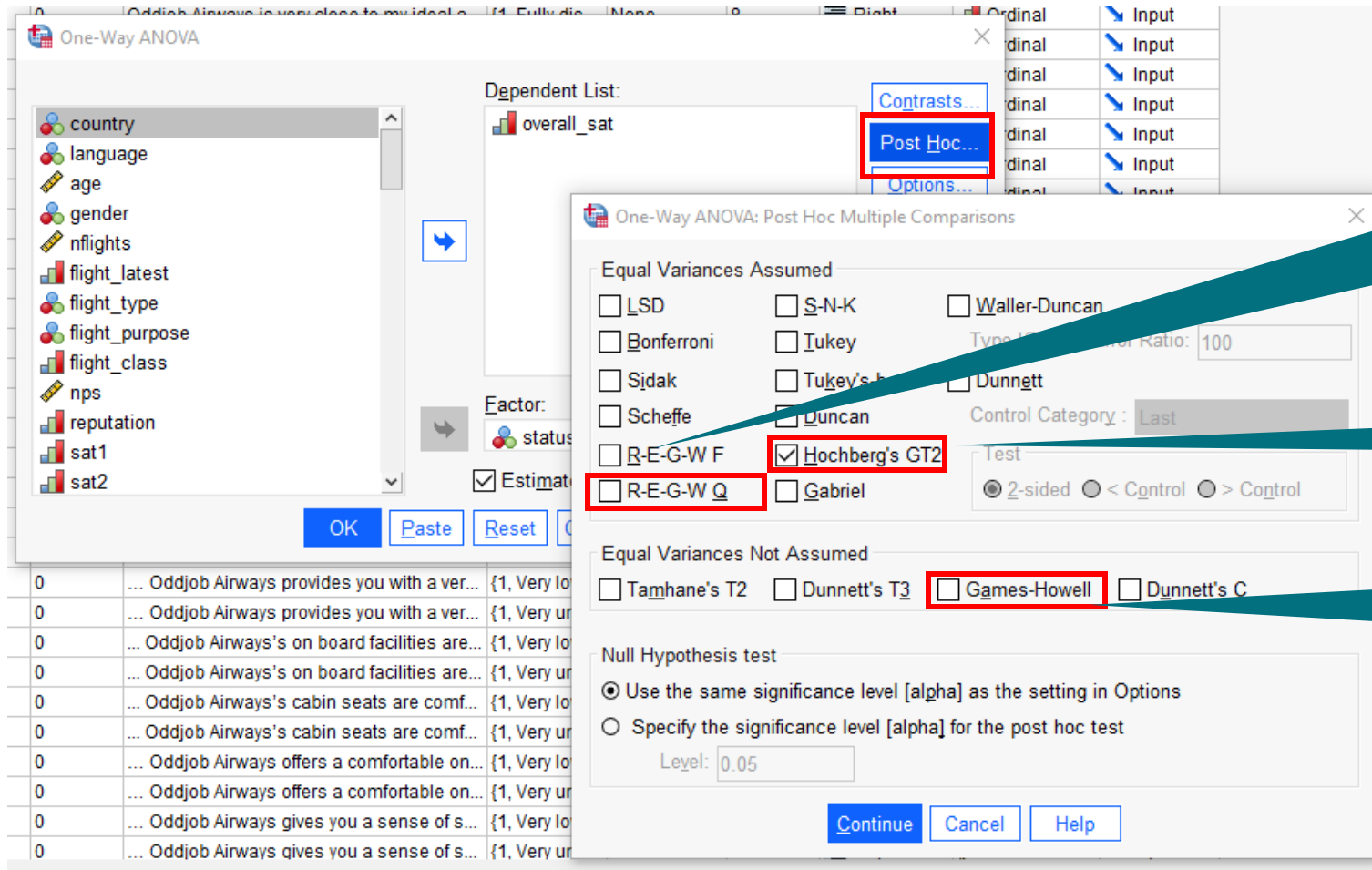
Overall, I am satisfied with the price performance ratio of Oddjob Airways.

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	51.755	2	25.878	9.963	<.001
Within Groups	2758.455	1062	2.597		
Total	2810.210	1064			

p-value ≤ 0.05
→ reject the null hypothesis that the mean of the overall price/performance satisfaction is the same between the status groups

One-way ANOVA

Analyze → Compare Means and Proportions One-way ANOVA → [Post Hoc] → Select the required statistics



Tick R-E-G-W Q
when you can
assume equal
variances but sample
sizes are equal

Tick Hochberg's GT2
when sample sizes
differ

Tick Games-Howell
when you cannot
assume equal
variances

One-way ANOVA

Multiple Comparisons

Dependent Variable: Overall, I am satisfied with the price performance ratio of Oddjob Airways.

Hochberg

(I) Traveler Status	(J) Traveler Status	Mean Difference (I-J)	Std. Error	Sig.	95% Confidence Interval	
					Lower Bound	Upper Bound
Blue	Silver	.440 [*]	.120	<.001	.15	.73
	Gold	.487 [*]	.148	.003	.13	.84
Silver	Blue	-.440 [*]	.120	<.001	-.73	-.15
	Gold	.047	.170	.990	-.36	.45
Gold	Blue	-.487 [*]	.148	.003	-.84	-.13
	Silver	-.047	.170	.990	-.45	.36

*. The mean difference is significant at the 0.05 level.

P-values < .05, therefore we conclude that *Silver* is different from *Blue* and that *Gold* is different from *Blue*.



One-way ANOVA

How to report the results of a one-way ANOVA test?

Example

In this study, we hypothesized that overall satisfaction with the price-performance ratio of Oddjob Airways differs based on traveler status. The results of a **one-way ANOVA test** indicate that traveler status affects overall satisfaction with the price-performance ($F(2, 1062) = 9.963, p < .001$). Therefore, our hypothesis was supported.

Further Hochberg post-hoc analyses reveal that blue-status customers are more satisfied than silver-status customers in terms of price-performance ratio ($M_{\text{blue}} = 4.47$ vs. $M_{\text{silver}} = 4.03, p < .001$). Additionally, blue-status customers express higher satisfaction compared to gold-status customers in the price-performance ratio ($M_{\text{blue}} = 4.47$ vs. $M_{\text{gold}} = 3.99, p < .01$). Other group comparisons do not yield significant differences.

F (between-groups degrees of freedom, within-groups degrees of freedom) = F value

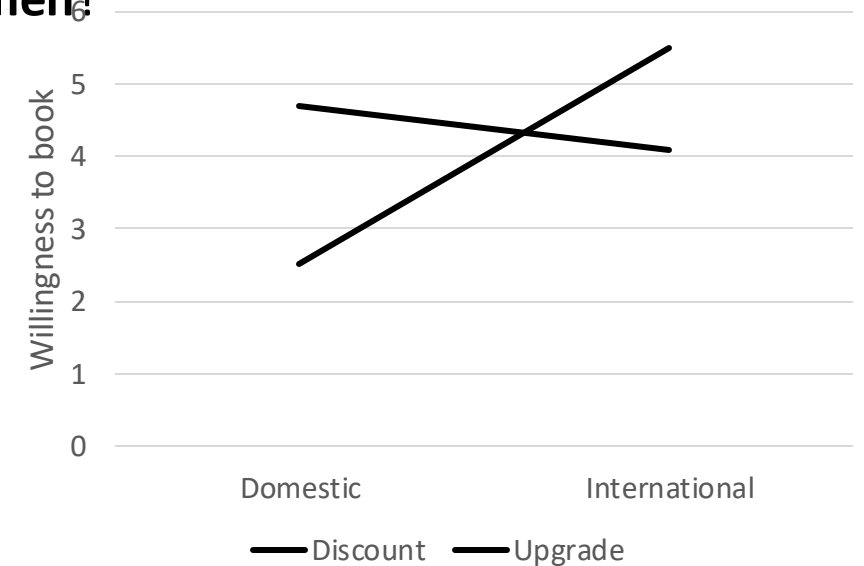


Two-Way ANOVA

- ❑ Compare the means of *three or more groups that are defined by 2 independent variables*
- Same assumptions as in One-way ANOVA
- Each factor's direct effect (also called main effect) on the dependent variable, as well as the factors' *interaction effect*.
- E.g., Does age influence satisfaction **differently for men and women?**

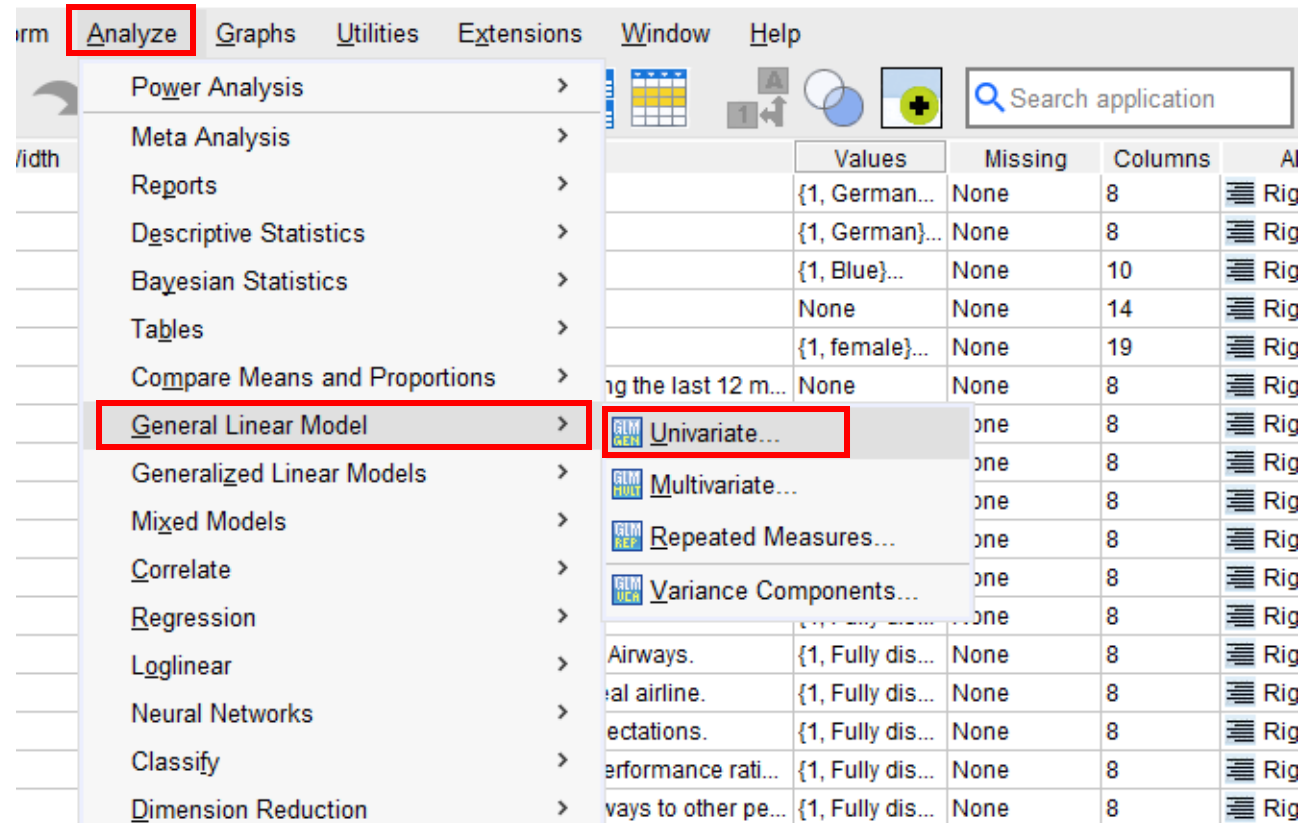
3 Null hypotheses

- There is no main effect of IV1 on the DV
- There is no main effect of IV2 on the DV
- There is no interaction effect between IV1 and IV2 on the DV



Two-Way ANOVA

Analyze → General Linear Model → Univariate → Select DV → Select 2 IVs (Fixed Factor(s)) → [Plots] → IV1 to Horizontal axis & IV2 to Separate Lines and 'add' → Continue → [options] → select Descriptive statistics & Homogeneity tests → continue





Two-Way ANOVA

Analyze → General Linear Model → Univariate → Select DV → Select 2 IVs (Fixed Factor(s)) → [Plots] → IV1 to Horizontal axis & IV2 to Separate Lines and 'add' → Continue → [options] → select Descriptive statistics & Homogeneity tests → continue

The image displays three SPSS dialog boxes for conducting a Univariate analysis, with red boxes and arrows highlighting the steps for setting up a Two-Way ANOVA.

Univariate Dialog Box:

- Dependent Variable:** s15 (highlighted with a red box).
- Fixed Factor(s):** flight_type and status (both highlighted with a red box).
- Options...** button (highlighted with a red box).

Univariate: Profile Plots Dialog Box:

- Factors:** flight_type and status.
- Horizontal Axis:** flight_type (indicated by a red arrow from the 'flight_type' factor).
- Separate Lines:** status (indicated by a red arrow from the 'status' factor).
- Add** button (highlighted with a red box).

Univariate: Options Dialog Box:

- Display:**
 - ☒ Descriptive statistics (highlighted with a red box).
 - ☒ Homogeneity tests (highlighted with a red box).

Red arrows indicate the flow of the process: from the 'Options...' button in the first dialog to the 'Add' button in the second, and then to the 'Descriptive statistics' and 'Homogeneity tests' checkboxes in the third.



Two-Way ANOVA

Descriptive Statistics

Dependent Variable: ... the food and beverage items served by Oddjob Airways are of a high quality.

What type was your latest flight?

	Traveler Status	Mean	Std. Deviation	N
Domestic	Blue	50.38	21.381	343
	Silver	53.53	22.239	133
	Gold	56.66	20.768	61
	Total	51.88	21.598	537
International	Blue	61.56	26.915	315
	Silver	54.75	23.857	106
	Gold	64.22	23.183	78
	Total	60.53	25.880	499
Total	Blue	55.73	24.808	658
	Silver	54.08	22.929	239
	Gold	60.90	22.396	139
	Total	56.04	24.136	1036

Levene's Test of Equality of Error Variances^{a,b}

		Levene Statistic	df1	df2	Sig.
... the food and beverage items served by Oddjob Airways are of a high quality.	Based on Mean	8.475	5	1030	<.001
	Based on Median	8.140	5	1030	<.001
	Based on Median and with adjusted df	8.140	5	1020.622	<.001
	Based on trimmed mean	8.632	5	1030	<.001

Tests the null hypothesis that the error variance of the dependent variable is equal across groups.

a. Dependent variable: ... the food and beverage items served by Oddjob Airways are of a high quality.

b. Design: Intercept + flight_type + status + flight_type * status

* Equal variances *not* assumed

Tests of Between-Subjects Effects

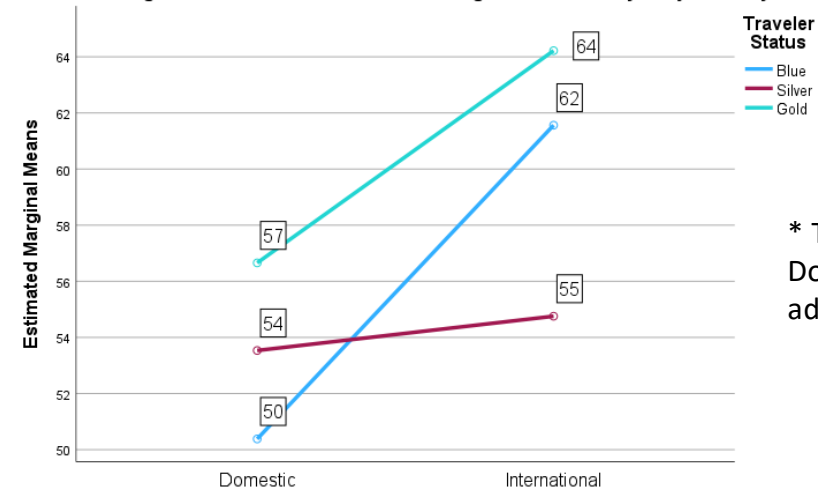
Dependent Variable: ... the food and beverage items served by Oddjob Airways are of a high quality.

Source	Type III Sum of Squares	df	Mean Square	F	Sig.
Corrected Model	26835.661 ^a	5	5367.132	9.596	<.001
Intercept	2226569.867	1	2226569.867	3980.764	<.001
flight_type	7626.272	1	7626.272	13.635	<.001
status	3486.227	2	1743.114	3.116	.045
flight_type * status	4333.152	2	2166.576	3.874	.021
Error	576112.297	1030	559.332		
Total	3856998.000	1036			
Corrected Total	602947.958	1035			

a. R Squared = .045 (Adjusted R Squared = .040)

both main effects and interaction effect significant

Estimated Marginal Means of ... the food and beverage items served by Oddjob Airways are of a h



* Tip
Double click the graph and add data labels using



Recap; Testing for Differences

Type of Tests	Purpose of Test	In SPSS
One Sample T-Test	Compare the mean of a question answered by one group of subjects against a constant on metric data	Analyze → Compare Means
Paired Samples T-Test	Compare two means of two different questions answered by the same group of subjects on metric data	Analyze → Compare Means
Independent Samples T-Test	Compare the mean of a question answered by two different groups of subjects on metric data	Analyze → Compare Means
ANOVA	Compare the means of a question answered by more than two different groups on metric data	One-way: Analyze → Compare Means Two-way: Analyze → General Linear Model → Univariate